TETANUS AND VACCINATION.[1] — AN ANALYTICAL STUDY OF NINETY-FIVE CASES OF THIS RARE COMPLICATION.

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During the year 1901 the occurrence of an unusual number of cases of small-pox in different parts of the United States stimulated the health authorities to a vigorous endeavor to protect the public by vaccination. Accordingly, during the year 1901, throughout many parts of the United States an unusual activity in vaccination took place.

The occurrence of a number of cases of tetanus succeeding vaccination, first in Cleveland, Ohio, and then in Camden, New Jersey, early attracted my attention, as this, to me, unknown complication seemed a matter of the greatest importance, increasing the danger of vaccination and correspondingly arousing the animosity of those misguided persons who have banded themselves together for the organized opposition of this well-recognized and only safeguard against small-pox.

The cases of tetanus in Camden have been so exploited in the newspapers and have been made the subject of so many editorials and comments in the medical press that they will, no doubt, form the starting point of numerous future attacks against vaccination, so that I deemed it important that as many cases of the complication as could possibly be collected be brought together and carefully analyzed, in order to determine whether tetanus be a necessary or an avoidable complication of vaccination. If necessary, we should be prepared for it and know how often to expect it; if avoidable, we should seek to eliminate it by every precaution regarding the selection of a superior virus and the performance of a careful operation.

[1] Read at the Second Annual Meeting of the American Association of Pathologists and Bacteriologists, March 28, 1902.

Tetanus is not a recognized Complication of Vaccination.— The only text-book I have been able to find make any mention of it is the recent edition of Osler's "Text-book of Medicine," in which it is simply stated that tetanus occasionally follows vaccination. In the "Minority Report" of the British Commission published in 1896, in which the disadvantages of vaccination are carefully and forcibly summarized, a single case, apparently the only one they were able to collect, is mentioned. Ordinary writers upon vaccination entirely ignore tetanus as a complication. The literature on the subject is extremely meager. In the Index Catalogue of the Library of the Surgeon General, but *seven* cases are mentioned under titles by which they can be recognized, and a total of fourteen cases are to be found in the literature. These occurred at somewhat remote periods in different parts of the country and were all attributed to secondary infection of the wound. It would seem reasonable, therefore, to conclude that tetanus has not been a frequent and important complication of vaccination in the past, either in this or any other country. It may, however, be surmised that what is true at the present time has also been true in the past, and that when tetanus occurred an attempt was made to suppress rather than publish it; but if in the past any such occurrence of tetanus had taken place as we have experienced recently, the medical literature would certainly contain some reference to it, as it does at present to the recent cases. For, suppressed as they are, very few cases having been published in detail, rumors of cases and editorials referring to cases and speculating upon their occurrence are to be found in many of the journals. The lack of published information concerning the complication cannot depend upon the failure to recognize it, as tetanus has been a well recognized disease for centuries, and indeed very little has been added to its symptomatology since the days of Hippocrates.

I have been able to collect a total of ninety-five cases. Throughout the paper I shall refer to *cases*, meaning by the term occurrences of tetanus that I have succeeded in authenticating and of which I have considerable detail, while the term "*rumor*" will be used for cases of tetanus the occurrence of which seems certain, but concerning which, for various reasons, I have been unable to secure the details.

TABLE I.

Cases from the literature		14
Cases collected		
a. With complete details	40	
b. With incomplete details	13	
	—	53
Rumors		28
		—
		95

Fatalities	61
Recoveries	24
Unknown	10
	—
	95

Adults	25
Children	45
Unknown	25
	—
	95

Males	38
Females	29
Unknown	28
	—
	95

TABLE II.

Tetanus Cases — Showing Chronological Occurrence.

The following table shows the chronological order of the cases I have succeeded in collecting:

Chronology.	Cases.	Rumors.
1854	1	
1878	1	
1882	3	
1886	1	
1889	1	
1891	1	
1892	0	6
1893	1	
1897	1	
1898	3	2
1899	3	
1900	1	
1901	45	18 = 63
1902	5	1
Unknown date		1
	—	—
	67	28
	28	67
	—	—
	95	95

January	0
February	0
March	1
April	0
May	0
June	1
July	1
August	1
September	1
October	1
November	24
December	7
Unknown	26
	—
	63

The large number of cases that occurred in the year 1901 indicates that some exceptional condition existed that changed an unimportant and infrequent complication into a very important and frequent one.

The occurrence of tetanus as a complication of vaccination has been variously explained as follows:

1. *That it is an accidental secondary infection of the vaccination sore.* Those holding this view think it depends upon conditions ever present and only to be expected, and that its occurrence is deplorable and preventable by the exercise of greater skill and caution in the performance of the operation

and the subsequent treatment of the wound. That such a microörganism as the tetanus bacillus, whose natural habitat is the soil and which occurs widely disseminated in Nature, might occasionally be able accidentally to find its way into vaccination lesions, cannot be gainsaid. It is certainly possible, and may occasionally happen, but there are very cogent reasons opposed to this view, and to content one's self with such a simple explanation may be to fall into egregious error, for if tetanus can thus occur, it should do so in all parts of the world, with more or less regularity.

The various indices to the medical literature are without any references to cases of tetanus following vaccination in the continental countries. A communication from the Imperial Health Office of Berlin informs me that the complication is *unknown in the German Empire.* Letters from the Pasteur Institute at Paris, the Pasteur Institute at Lille, and the French Institute inform me that the complication is *unknown in France.* No cases appear to occur in Italy, Russia, Austria, or other European countries, so that we are engaged in the consideration of a complication that is chiefly American, has become important within a year, cannot be reasonably attributed to geographic, telluric, atmospheric, or social conditions, but must depend upon some mode of operating, some mode of preparing the virus, or some mode of treating the wound in vogue at the present time, but not in the past.

2. *That the occurrence of the complication depends upon a local prevalence of the tetanus bacilli.* The epidemic in Camden has been attributed to germs in the dust of that city, supposed to depend upon a prolonged period of dry weather. This entirely fails to explain the matter, for were this the true solution we should find ordinary traumatic tetanus occurring more frequently than usual. This is, however, not the case, as aside from the vaccination cases there were fewer than the usual number of cases both in Camden and Philadelphia. Further, while it is true that a greater number of cases occurred in Camden than in other places of equal size, the occurrence of tetanus following vaccination has too

wide a geographical distribution to be explained in this way.

TABLE III.

Report of the Deaths from Tetanus in Camden City for the following Years.

	1901	1900	1899	1898	1897	1896	1895	1894	1893	1892	1891	1890
January	....	....	....	....	....	....	....	....	....	....	....	....
February	....	....	....	....	....	....	....	....	....	....	1	....
March	....	....	....	....	....	....	....	....	....	....	....	....
April	....	....	....	....	....	....	....	....	....	....	....	....
May	....	....	1	....	1	....	....	....	....	....	....	....
June	....	....	....	....	....	....	1	....	....	....	....	....
July	1	....	....	....	....	....	....	....	....	....	....	....
August	....	....	....	2	....	....	....	....	....	....	....	....
September	....	....	....	1	....	....	....	1	....	....	....	1
October	....	....	....	....	....	....	....	....	....	...	....	....
November	9	....	....	....	....	....	....	....	....	....	....	....
December	1	1	....	....	....	....	....	....	....	....	....	....

Cases are reported from seventeen different States in the United States, from the Dominion of Canada, Porto Rico, the Philippine Islands, Cuba, etc.

That geographical conditions may exert a pronounced influence upon the occurrence of the tetanus, tetanus being known to be particularly prevalent in certain districts. Thus it is said that Cuba and Porto Rico have soils particularly rich in tetanus bacilli, and that Long Island has a similar dangerous soil. That such an influence is important is, however, very doubtful, as from these territories we have very few cases reported. (3 Cuba, 3 Porto Rico, 2 Long Island.)

Table IV.

Tetanus Cases — Showing the Geographical Distribution.

	Cases.	Rumors.
Canada.		
Three Rivers, Quebec	1	
St. John, N.B.	1	
Connecticut.		
South Glastonbury	1	
Cuba.		
Havana	3	
England.		
Lancet	1	
Commissioners' Report	1	
Illinois.		
Chicago	0	1
Louisiana.		
New Orleans	1	
Maine.		
Biddeford	1	
Maryland.		
Baltimore	1	
Massachusetts.		
Belmont	1	
Cambridgeport	1	
East Dennis	1	
Boston	1	
Michigan.		
Kalamazoo	1	
Minnesota.		
Minneapolis	1	
Taylors Falls	1	
Missouri.		
St. Louis	0	1
New Jersey.		
Atlantic City	4	2
Bridgeton	1	
Camden	11	

	Cases.	Rumors.
New Jersey.		
Jordantown	0	1
Millville	1	
New York.		
Albany	1	
Long Island	2	
Oswego	1	
Ocean.		
"City of Para"	1	
Ohio.		
Cincinnati	0	1
Cleveland	4	
Pennsylvania.		
Bristol	2	
Easton	1	
Girardville	1	
Lyndell	1	
Millbach	0	1
Philadelphia	11	12
Rosemont	0	1
Philippine Islands	1	
Porto Rico	1	2
Scotland	1	
South Carolina.		
Charleston	1	
Tennessee.		
Paris	1	
Unknown	0	6
Virginia.		
Richmond	1	
Wisconsin.		
Milwaukee	1	
	67	28
	28	67
	95	95

3. *That carelessness in the treatment of the vaccination wound is the source of the difficulty, because it facilitates accidental secondary infection.* While this has a ring of genuineness about it, the argument is extremely unconvincing, if one take into consideration the fact that for one hundred years vaccinations have been performed with a total disregard to cleanliness and asepsis in all parts of the world, on all classes of people, in towns and cities, by careless and careful physicians, upon clean and dirty persons, and that it has been unusual for any dressing to be applied in the past, and that during these many years tetanus has remained almost an unknown complication. We cannot conceive of any difference between the social conditions at the present and those of the past that are not immensely in favor of the present; yet it is at present that we find tetanus.

In our own country greater care is undoubtedly exercised at the present time than heretofore, and in those cases in which, because of ignorance and poverty, the same conditions prevail at the present time that existed a hundred years ago, as for example among the peasantry of Europe, we find tetanus practically unknown.

Vaccination wounds are treated with much greater care at the present time than ever before. It is but now becoming recognized that vaccination is an operation, and that to free it from the common dangers of all operative manipulations, cleanliness and care are essential. We, therefore, find that at present the skin is cleansed and disinfected and the lesion itself protected and subsequently dressed with a care never dreamed of a few years ago; yet when we come to examine the details of those cases which have come within our knowledge, we find that this care of the wound appears to be without influence upon the development of tetanus, for while many cases have occurred among ignorant and filthy children, in an equally great number of cases not only ordinary but extraordinary care seems to have been exercised. Thus one case occurred in the person of an adult sister of a physician in Cleveland, Ohio. Both the patient and the operator were refined and cultured people, were apprehensive of the results, and exerted unusual and extreme precautions, yet despite all, death from tetanus followed this vaccination. Here let it be said that the virus apparently responsible for so many cases was used.

The densely ignorant and filthy people of the island of Porto Rico, with no knowledge of personal hygiene, living in a place reputed to be extremely dangerous because of tetanus, were vaccinated by the United States authorities after the occupation of that territory, and out of some eight hundred thousand vaccinations three cases of tetanus, two of which are very doubtful, are reported by Dr. George G. Groff.

4. *The use of a shield to protect the vaccination wound has recently become quite common, and has been blamed for the occurrence of tetanus.* It has, however, met with violent and perhaps justifiable condemnation, on the ground that the pressure of its edges and of the adhesive plaster bandages by which it is held in place obstructs the lymphatic circulation and increases the severity of the wound and the danger of infection, also that the shield induces anaerobic conditions. Upon looking over the cases reported, however, we find so few in which shields were used that they seem to have had no influence upon the occurrence of tetanus.

The Relation of Tetanus to the Vaccine Virus.

With the evolution of vaccination as a prophylactic measure, certain changes in technique have been gradually introduced. Thus bovine virus has now displaced the arm to arm and human scab vaccinations previously employed, and within the last quarter-century the use of the human virus has been almost entirely given up. Can it be that the danger of tetanus has something to do with the employment of bovine virus? Only a few of the early cases drawn from the literature followed the use of human virus. This seems possible when we consider that it is only in the last half century that any cases have occurred, but can scarcely be, inasmuch as bovine virus is the only form employed in Belgium, Germany, France, and other European countries, yet we find no tetanus there, though it is common in our own country at the present time. While it is suggestive that the first tetanus cases reported in the literature make their appearance about the time bovine virus came into general use, it is not true that the number of cases increased in proportion to the popularity of the bovine virus; for from 1854 when the first case makes its appearance in the literature until 1901, only isolated cases usually at intervals of years are reported, a sudden extraordinary increase being observed in 1901.

The next step in the evolution of vaccination was the improvement in the quality of the virus, suggested by Copeman in 1891, by which glycerine was permitted to act for some time upon the virus, for the purpose of destroying microörganisms which it contained. By the use of the glycerine, a bacteria-free virus can be secured. This improved virus made slow but steady progress, until at the present time it has attained to the greatest popularity and bids fair to replace all other forms. It is used almost exclusively on the continent of Europe and has made very large inroads into this country.

The glycerinized virus is so new, having been in use but ten years and in common use little more than five years, that we are scarcely able to express a positive opinion regarding its advantages and disadvantages. The advantage of having the contaminating bacteria destroyed by the glycerine is indisputable; the disadvantage of occasionally having the glycerine act so long that the specific germs of vaccinia are killed and the virus made inert, is also indisputable. That at times when the demands upon the manufacturer are exceptionally great, the virus mixed with the glycerine might be placed upon the market before the necessary time had elapsed for the contained bacteria to be killed, is a somewhat grave danger. These possibilities have somewhat lessened the confidence with which the preparation is received and used by many practitioners.

When we come to investigate the cases of tetanus concerning which information is at hand, we are somewhat startled to find that a large number of them have succeeded the employment of this supposedly best and most refined preparation. This point will be discussed below.

TABLE V.

Showing Relation of Tetanus to Make of Virus.

	Dry.	Glycerinized.		
		Points.	Tubes.	
E.	3	10	17 =	30
A.	2		=	2
D.	0		2 =	2
W.	0	1	=	1
S.	0		3 =	3
I.	1		=	1
G.	1		=	1
	7	11	22	40 cases.

Relationship to Particular Brands of Virus.

When an attempt is made to determine whether any particular brand or make of virus yields an unusually large number of tetanus cases, we are confronted by a still more suggestive fact; that is, that while an occasional case of tetanus has succeeded the use of the virus of nearly all of the large manufacturers, the great majority of the cases have succeeded the use of a particular brand of virus.

The brands of virus to be considered are called E., D., A., S., W., I., St., and Wd. Those called E., A., and D. are the products of the largest manufacturers, and it is not improbable that they do about an equal amount of business. It is probably safe to say that these vaccines are all made under careful conditions, with the exercise of as great an amount of skill as can be exerted with our present knowledge. No care or expense is spared to have these products perfect. We find, however, a remarkable discrepancy in the results following their employment — a discrepancy that leads me to conclude that tetanus bacilli may be contained in the virus and distributed with it.

The occurrence of thirty cases of tetanus after the employment of virus E. indicates that in it tetanus germs were present in larger proportions than in any other, though the occurrence of only thirty cases following the use of millions of doses indicates that the number of tetanus organisms present was small.

The Occurrence of the Cases in Groups.

In this connection must be pointed out as interesting and significant the groups of cases that have occurred from time to time. Thus in Cleveland, O., during 1901, we note the occurrence of four cases. In Camden, N.J., during October, November, and December, 1901, the occurrence of eleven cases. At Atlantic City, about the same time, five cases, and in Philadelphia and vicinity nearly at the same time about twenty-five cases. If we analyze these cases we find one vaccine (virus E.) chiefly if not exclusively implicated. The most instructive group of cases that I have been able to study occurred in the Philadelphia Hospital. In this institution with nearly four thousand five hundred inmates, there was a threatened epidemic of small-pox depending upon the admission of a case of small-pox from the street. After one or two cases had developed within the institution, it was decided to vaccinate every inmate, and the resident physicians went systematically through the institution vaccinating sick and well alike. The institution was thus vaccinated with the exception of the Men's Insane Department, the greater number of whose inmates were obliged to wait a few days until a new consignment of virus arrived. With this new consignment (virus E.) they were then all vaccinated. Upon looking up the statistics of the hospital, we find that in the insane departments, male and female, no case of spontaneous traumatic tetanus had developed within twelve years. It is, in fact, not known that there ever has been a case of traumatic tetanus developed within the walls of the institution, but this cannot be determined, as the records prior to twelve years ago were destroyed by fire. Succeeding the vaccination, however, a group of tetanus cases, confined exclusively to the Men's Insane Department, occurred. Here five typical cases with trismus and opisthotonos and every marked symptom of the disease occurred, all being followed by death, four from trismus itself, and one from pneumonia, occurring immediately after a cessation of the spasms. The occurrence of this outbreak occasioned much alarm, so that every suspicious vaccination wound observed was thoroughly excised and treated antiseptically. After this excision of the wounds, *eleven additional cases developed* trismus and muscular rigidity, though after the administration of enormous doses of antitoxin they all recovered. In going carefully over the details of these cases, I find that with one very doubtful exception, every patient that developed tetanus was vaccinated with the same virus which had caused tetanus at Cleveland, Camden, Atlantic City, Philadelphia, and elsewhere (virus E.).

Should it be suggested that the occurrence of groups of tetanus cases depends upon the popularity of the virus in certain districts, as Philadelphia, Camden, Atlantic City, where it was almost exclusively used, and that in these districts some telluric, atmospheric, or other condition prevailed, causing them to be more predisposed to tetanus than the country in general, the idea should be at once dispelled by a few moments' consideration of the statistics presented. Thus if we deduct from the total thirty cases attributable to virus E., the six positive cases occurring in Camden, the seven positive cases occurring in Philadelphia, and the three positive cases occurring in Cleveland, there remain against this virus a total of fourteen cases, which is greater than the sum total of the cases referable to all the other viruses produced in the country, thus showing that even where the cases are scattered, and not in groups, this virus E. has four times as many cases to its credit as any other one virus and many more than all the other viruses put together.

Thus convinced that the bacilli which cause tetanus subsequent to vaccination are in the virus, we are obliged to prove our position, which we do by a statistical study of the material collected. (Refer to Table IV.)

Forty cases are presented with exact information concerning the make and form of virus employed. Of these forty cases *thirty* follow the employment of virus E., *ten* follow the use of all other forms combined, and no other single virus has a higher number than *three*. This seems to show quite convincingly that there is something about virus E. that is different from the others. We have in addition to these forty cases with complete details, eight cases in which it is known positively that one or the other of two viruses was used, but in which it cannot now be certainly determined which of the two it was.

TABLE VI.

Cases of Disputed Virus where Names are Given.

Cases, No.[1]	E.	A.	D.	W.	Wd.	St.
25	1	. .	. .	1	. .	. .
29	1	. .	1	. .	. .	. .
45	1	. .	. .	. .	. .	1
56	1	1	. .	. .	. .	. .
71	1	1	. .	. .	. .	. .
72	1	1	. .	. .	. .	. .
73	1	1	. .	. .	. .	. .
91	. .	. .	1	. .	1	. .
	7	4	2	1	1	1

We find that in the cases of this group vaccine E. figures in seven. If we assume, which it would be fairly justifiable to do, that all of these cases belong to E., then the disproportion is changed from

E., 30: all others 10; to
E., 37: all others 11;

thus making matters much worse. If, however, we proceed on the reverse order and admit that E. may be erroneously charged with these cases, we find the proportion

E., 30: all others 18,

so that matters are but little improved for virus E.

[1] The numbers refer to my list of cases, which cannot be published in detail, because much that it contains is confidential.

TABLE VII.

Table of Comparisons.

Viruses.	Known.	Disputed.
E.	30	7
A.	2	4
S.	3	0
D.	2	2
W.	1	1
Wd.	0	1
St.	0	1
G.	1	0
I.	1	0

Dry and Glycerinized Viruses.

The respective influences of dry and glycerinized virus upon the causation of tetanus is shown by reference to Table V., where we see that seven cases followed the use of dry points, eleven the use of glycerinized points, and twenty-two the glycerinized tube virus. This proportion, seven to thirty-three, appears to be convincingly in favor of the dry virus, but we find an error resulting from the use of virus E. All three forms of virus E. have been followed by tetanus in the respective proportions of dry points three, glycerinized points ten, and glycerinized tube virus seventeen. If we entirely omit virus E. from the consideration, we find the totals of all other viruses to be;

Dry points,		4
Glycerinized points,	1	$\left.\right\} = 6$
Glycerinized tubes,	5	

The proportion of four dry to six glycerinized being about correct when we remember that the glycerinized is the popular virus at the present time.

This seems to show pretty well that there is no disproportion in favor of dry over glycerinized virus as regards the occurrence of tetanus.

The Source of the Tetanus Bacilli in Virus.

When we consider the distribution of the tetanus bacillus in nature, we find it in the soil, chiefly where it has been well fertilized. The tetanus bacillus is, no doubt, frequently swallowed by herbivorous animals in browsing upon the surface of the ground. The anaerobic conditions for its growth being excellent, we find that the intestines of herbivorous animals commonly contain large numbers of the bacilli, and that with the intestinal evacuations, the bacteria are deposited again upon the soil in increased numbers. The first source of danger, therefore, by which vaccine virus can be contaminated is the manure of the calf, the next source the dusts arising from the dry manure and the soil, both of which may be brought into the stables with the hay, or upon the animals.

When large hay fed animals are employed for the manufacture of virus, the manure is certainly a source of danger, but in stables where sucking calves are used for the purpose, fewer tetanus bacilli are present in the excrement. However, Huddleson found them present in the feces of eight per cent of the small calves used in the laboratory of the Health Department of New York.

It is evident that from these sources opportunities occur for the entrance of tetanus germs into the vaccine virus, but as a matter of fact no one has yet succeeded in finding them in the virus. This indicates that their number must be so small that in the distribution of the virus in tubes and on points, hundreds of thousands of tubes escape where one tube is contaminated. From these data we have been forced to conclude that it is unusual for many tetanus germs to be implanted at the time of vaccination, but that such implantations can and do occur.

The Incubation Period.

The period of incubation of ordinary traumatic tetanus varies from a few hours to several weeks (see Plate XL., Chart A), according to statistics derived from the study of a large number of cases.

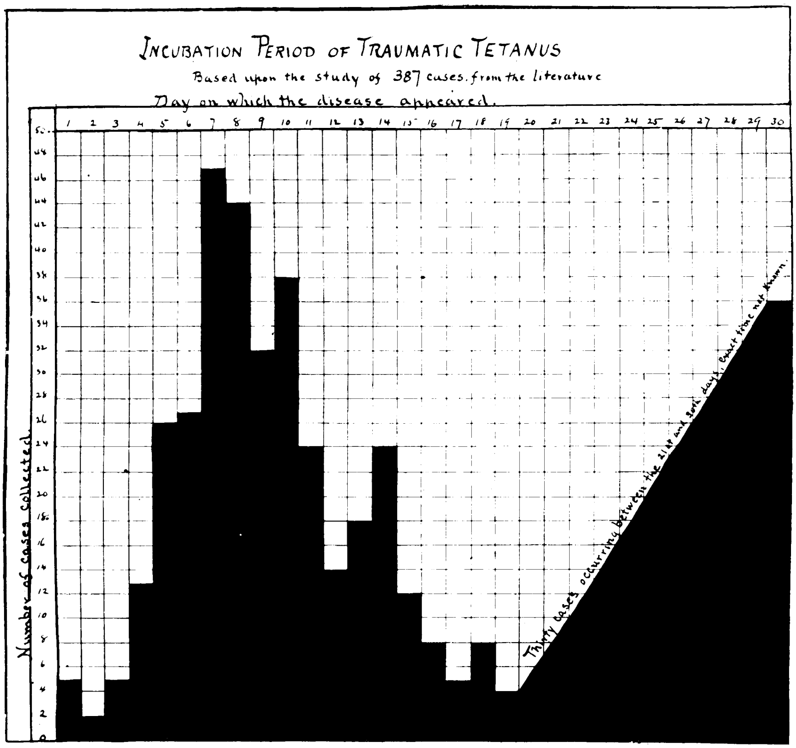

CHART A.

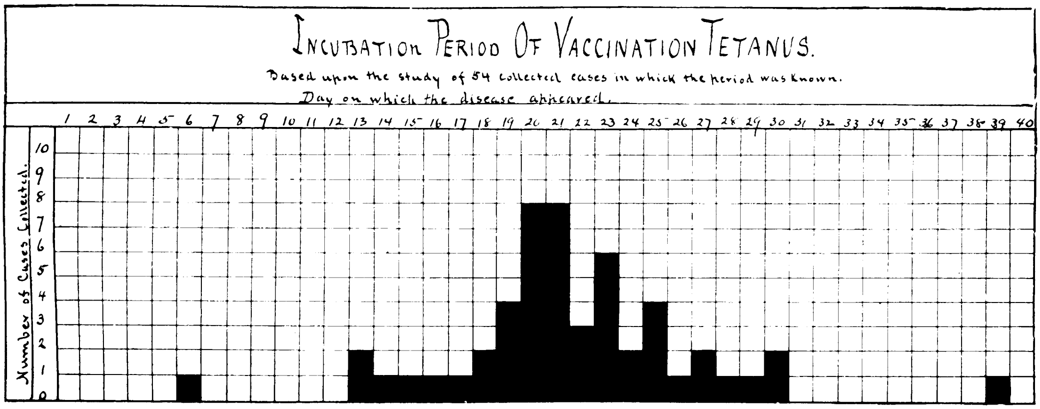

CHART B.

TABLE VIII.

Tetanus Cases — Showing Incubation Period.

Shortest period	6	days.
Longest period	39	"
Average	22	"
1. For Points — Shortest period .	30	"
Longest period .	16	"
Average	23	"
2. For Glyc. Virus — Shortest period,	6	"
Longest period .	39	"
Average . .	22	"

The usual incubation period varies from seven to nine days. In the vaccination cases, that is, in cases of tetanus following vaccination, we find the shortest period of incubation to be six days, the longest thirty-nine, the average twenty-two days (Plate XL., Chart B). It is thus seen that tetanus following vaccination appears on the average about two weeks later than ordinary traumatic tetanus. This is the sole weakness in the argument, and I think if this error could be eliminated it would at once be conceded that the remaining facts are indisputable. The prolonged incubation period is usually interpreted to signify that the infecting organism entered the vaccination wound at a time when it was open, exposed, and subject to secondary infection from the air, water, clothing, etc., with which it comes in contact. In truth, however, how rarely do superficial ulcerations become infected with tetanus! Did any one ever hear of a sudden paroxysmal outbreak of sixty-one cases of tetanus in one year following the infection of leg-ulcers, furuncles, mosquito bites, etc., even when cleanliness was not observed?

I think after the statistical demonstration given, that this secondary infection theory must be looked upon as a misinterpretation. The probability is that the tetanus bacillus is ingrafted into the skin at the time of vaccination, but fails to find suitable conditions for its growth until after the development of the vaccine lesion paves the way by the local destruction of tissue. This makes us add to the usual incubation an additional period (a couple of weeks) during which the wound has been prepared for the implanted organisms, thus bringing the entire period to the length usually observed in the vaccination cases. An examination of Table VIII. shows that no difference exists between dry vaccine on points and glycerinized virus regarding the length of the incubation period.

In nearly every case of which I have succeeded in securing details, I learned that the vaccination "took well." This may not be without significance, though it may be interpreted in several ways. It may mean that those vaccinations in which severe local lesions occur are those in which secondary infection has the best chance of occurrence. If so, it means that such lesions are dangerous and that mild viruses should be used, the multiple insertion of a mild virus being preferable to a single insertion of an active virus. It may also mean that the lesion was caused by an impure virus, in which tetanus as well as other organisms might be contained. Lastly, and I think truly, it may mean that it is only when such local lesions occur that the implanted tetanus bacilli can find conditions suitable for their development.

Reference is again to be made to the chronological development of the cases, by which it will be observed that the majority of cases occur at a time when, because of agitation concerning small-pox, the demands for virus are great. This being the case, it is not impossible that vaccine viruses are sometimes marketed prior to the time when the action of the glycerine has extinguished the life of the bacteria in the virus.

Conclusions.

From the foregoing the following conclusions seem justifiable:

1. Tetanus is not a frequent complication of vaccination, a total of ninety-five cases having been collected.

2. The number of cases recently observed is out of all proportion to what has been observed heretofore.

3. The cases are chiefly American and occur scattered throughout the eastern United States and Canada.

4. They have nothing to do with atmospheric, telluric, or seasonal conditions.

5. They occur in small numbers after the use of various viruses.

6. An overwhelming proportion has occurred after the use of a particular virus.

7. The tetanus organism may be present in the virus in small numbers, being derived from the manure and hay.

8. Occasionally the number of bacilli becomes greater than usual through carelessness or accident.

9. The future avoidance of the complication is to be sought for in greater care in the preparation of the vaccine virus.

[Since writing this paper, the final proof of its correctness has been given by the discovery of tetanus bacilli in some of the same vaccine virus used at the time the chief outbreak occurred in 1901. This discovery was made by Dr. R. W. Wilson, and communicated to the Philadelphia County Medical Society on April 23, 1902.]